NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

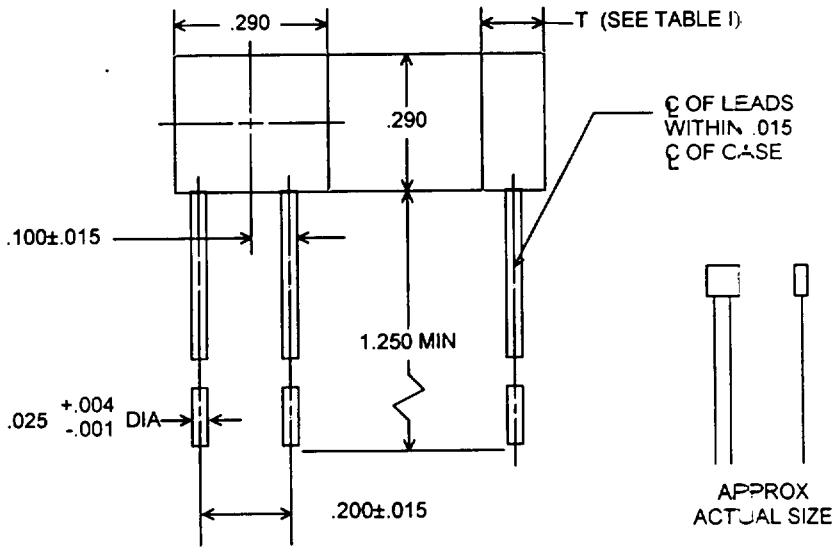
MIL-PRF-39014/20C
6 August 1985
SUPERSEDING
MIL-C-39014/20B
14 September 1984

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC,
(GENERAL PURPOSE),
ESTABLISHED RELIABILITY
STYLE CKR08

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-PRF-39014.



INCHES	MM	INCHES	MM
.001	0.03	.100	2.54
.004	0.10	.200	5.08
.015	0.38	.290	7.37
.025	0.64	1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .010 (0.25 mm).

FIGURE 1. Capacitor style CKR08.

© denotes changes

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:27Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Molded or preformed.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55°C to +125°C.

Voltage rating: See table I.

Dissipation factor: Not more than 2.5 percent.

Dielectric withstanding voltage:

Body insulation: Shall be able to withstand 1,300 volts between leads and case.

Insulation resistance:

Measured between one terminal and the case:
At 25°C: Not less than 100,000 megohms or 1,000 megohm-microfarads, whichever is less.
At 125°C: Not less than 10,000 megohms or 100 megohm-microfarads, whichever is less.

Insertion loss: Not applicable.

Seal: Not applicable.

Voltage-temperature limits: In accordance with MIL-C-39014.

Capacitance change with reference to 25°C	
Steps A through D of table V Bias = 0 volts	Steps E through G of table V Bias = rated voltage
±15 percent	+15, -25 percent

Immersion:

IR: Not less than 50 percent of initial 25°C requirements.

Salt spray (corrosion): In accordance with MIL-C-39014.

Ⓒ Resistance to soldering heat:

ΔC: +15, -5 percent of the initial 25°C measurement.

Life:

Rated conditions: 100 percent of dc rated voltage at 125°C.

0 through 3,000 hours:

IR: At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 2.5 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:27Z
Check the source to verify that this is the current version before use.

4,000 through 32,000 hours:

IR: At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 2.5 percent.

Accelerated conditions, 200 percent of dc rated voltage at 125°C.

0 and 250 hours:

IR: At 25°C, not less than 50 percent of initial 25°C requirement.
At 125°C, not less than 50 percent of initial 125°C requirement.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 2.5 percent.

1,000, 2,000 and 4,000 hours:

IR: At 25°C, not less than 15 percent of initial 25°C requirement.
At 125°C, not less than 15 percent of initial 125°C requirement.
ΔC: ±20 percent from initial measured value.
Dissipation factor: Not greater than 2.5 percent.

Part number: M39014/20- (dash number from table I).

TABLE I. Electrical characteristics and dash number.

Dash number				Capacitance	Capacitance tolerance	DC rated voltage	Dimension (in inches 1/)	
Failure rate level (%/1,000 hours)								
1.0 (M)	0.1 (P)	0.01 (R)	0.001 (S)					
				pF	± percent	volts		
0004	0007	0010	0013	1,200,000	10	50	.130 (3.30)	±.010 (.25)
0005	0008	0011	0014	1,500,000	10	50	.130 (3.30)	±.010 (.25)
0006	0009	0012	0015	2,000,000	10	50	.130 (3.30)	±.010 (.25)
0104	0107	0110	0113	1,200,000	10	50	.145 (3.68)	±.015 (.38)
0105	0108	0111	0114	1,500,000	10	50	.145 (3.68)	±.015 (.38)
0106	0109	0112	0115	2,000,000	10	50	.145 (3.68)	±.015 (.38)

1/ Metric equivalents are given in parentheses.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:27Z
Check the source to verify that this is the current version before use.

TABLE II. Substitutability data.

Dash numbers in MIL-C-39014/20B 1/				Substitutable for dash numbers in:									
				MIL-C-39014/20A(EC)					MIL-C-39014/20(EC)				
M	P	R	S	L	M	P	R	S	L	M	P	R	S
0004	0007	0010	0013	0001	0004	0007	0010	0013	0001	0004	0007	0010	0013
0005	0008	0011	0014	0002	0005	0008	0011	0014	0002	0005	0008	0011	0014
0006	0009	0012	0015	0003	0006	0009	0012	0015	0003	0006	0009	0012	0015
0104	0107	0110	0113	0101	0104	0107	0110	0113	---	---	---	---	---
0105	0108	0111	0114	0102	0105	0108	0111	0114	---	---	---	---	---
0106	0109	0112	0115	0103	0106	0109	0112	0115	---	---	---	---	---

1/ Dash numbers in lower FR level columns are substitutable for dash numbers in all higher FR level columns.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Navy - OS, TD
DLA - ES

User activities:
Navy - AS, SH, MC

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5910-1542-9)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:27Z
Check the source to verify that this is the current version before use.